

# Time-like and space-like electromagnetic form factors of nucleons, a unified description

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## Nucleon Electromagnetic Form Factors

- Definition and properties

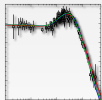


## The Lomon-Gari-Krümpelmann model



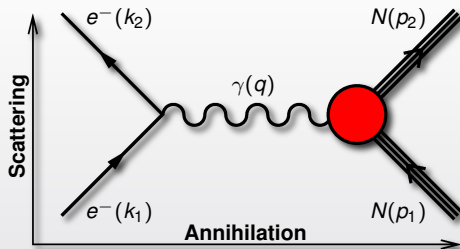
## Time-like extension of the model

- Breit-Wigner regularization procedure



## Data, fits, results and discussion

# Dirac and Pauli Nucleon Form Factors



Scattering:  $e^- N \rightarrow e^- N$   
**Space-like** kinematic region

$$q^2 = -2\omega_1\omega_2(1 - \cos \theta_e) \leq 0$$

Annihilation:  $e^+ e^- \leftrightarrow N\bar{N}$   
**Time-like** kinematic region

$$q^2 = 4\omega^2 > 0$$

Scattering amplitude  
in Born approximation

$$\mathcal{M} = \frac{1}{q^2} [e \bar{u}(k_2) \gamma_\mu u(k_1)] \underbrace{[e \bar{U}(p_2) \Gamma^\mu(p_1, p_2) U(p_1)]}_{\text{Nucleon EM 4-current: } J_N^\mu}$$

**From Lorenz and gauge invariance**

$$\Gamma^\mu(p_1, p_2) = \gamma^\mu F_1^N(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2M_N} F_2^N(q^2)$$

Dirac FF

$$F_1^N(q^2)$$

- Non-helicity flip
- $F_1^N(0) = \mathcal{Q}_N$

Pauli FF

$$F_2^N(q^2)$$

- Helicity flip
- $F_2^N(0) = \kappa_N$

$\mathcal{Q}_N = N$  electric charge

$\kappa_N = N$  anomalous magnetic moment



# Sachs Nucleon Form Factors

## Breit frame

No energy exchanged

$$p_1 = (E, -\vec{q}/2)$$

$$p_2 = (E, \vec{q}/2)$$

$$q = (0, \vec{q})$$

## Nucleon EM 4-current

$$J_N^\mu = (J_N^0, \vec{J}_N) \quad \left\{ \begin{array}{l} \rho_q = J_N^0 = e \left[ F_1^N + \frac{q^2}{4M_N^2} F_2^N \right] \\ \vec{J}_N = e \bar{U}(p_2) \vec{\gamma} U(p_1) \left[ F_1^N + F_2^N \right] \end{array} \right.$$

## Sachs Nucleon Form Factors

$$G_M^N(q^2) = F_1^N(q^2) + F_2^N(q^2) \quad G_E^N(q^2) = F_1^N(q^2) + \frac{q^2}{4M_N^2} F_2^N(q^2)$$

In the Breit frame represent the **Fourier transforms** of **charge** and **magnetic moment spatial distributions** of the nucleon

Normalization at  $q^2 = 0$

- $G_E^N(0) = Q_N$

- $G_M^N(0) = \mu_N$

$\mu_N = Q_N + \kappa_N$  is the nucleon magnetic moment

## Isospin decomposition

Isoscalar components

- $F_{1,2}^{\text{is}} = F_{1,2}^p + F_{1,2}^n$

- $G_{E,M}^{\text{is}} = G_{E,M}^p + G_{E,M}^n$

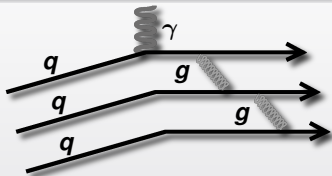
Isvector components

- $F_{1,2}^{\text{iv}} = F_{1,2}^p - F_{1,2}^n$

- $G_{E,M}^{\text{iv}} = G_{E,M}^p - G_{E,M}^n$

# pQCD asymptotic behavior

## Space-like region



- **pQCD:** as  $q^2 \rightarrow -\infty$ ,  $F_1(q^2)$  and  $F_2(q^2)$  must follow counting rules
- Quarks exchange gluons to distribute momentum transferred by the photon

### Dirac form factor $F_1$

- Non-helicity flip
- Two gluon propagators
- $F_1(q^2) \underset{q^2 \rightarrow -\infty}{\sim} (-\tilde{q}^2)^{-2}$

### Pauli form factor $F_2$

- Helicity flip
- Two gluon propagators
- $F_2(q^2) \underset{q^2 \rightarrow -\infty}{\sim} (-\tilde{q}^2)^{-3}$

### Sachs form factors $G_E$ and $G_M$

- $G_{E,M}(q^2) \underset{q^2 \rightarrow -\infty}{\sim} (-\tilde{q}^2)^{-2}$
- Ratio:  $\frac{G_E}{G_M} \underset{q^2 \rightarrow -\infty}{\sim} \text{constant}$

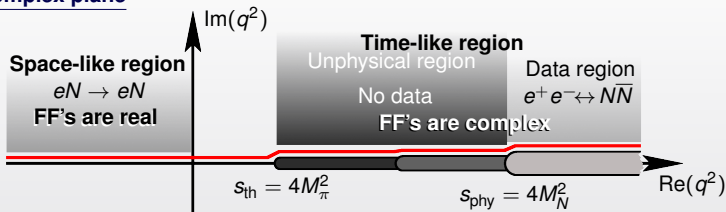
### QCD correction

$$\tilde{q}^2 = q^2 \ln \left( \frac{-q^2}{\Lambda_{\text{QCD}}} \right)$$



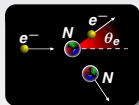
# Cross sections and analyticity

## $q^2$ -complex plane



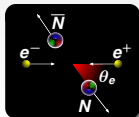
Crossing: tot. helicity =  $\begin{cases} 1 \Rightarrow G_E \\ 0 \Rightarrow G_M \end{cases}$

$G_E(4M_N^2) = G_M(4M_N^2)$



## Elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 \omega_2 \cos^2 \frac{\theta_e}{2}}{4\omega_1^3 \sin^4 \frac{\theta_e}{2}} \left[ G_E^2 - \tau \left( 1 + 2(1-\tau) \tan^2 \frac{\theta_e}{2} \right) G_M^2 \right] \frac{1}{1-\tau} \quad \tau = \frac{q^2}{4M_N^2}$$



## Annihilation

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 \beta C}{4q^2} \left[ (1 + \cos^2 \theta) |G_M|^2 + \frac{1}{\tau} \sin^2 \theta |G_E|^2 \right]$$

$\beta = \sqrt{1 - \frac{1}{\tau}}$



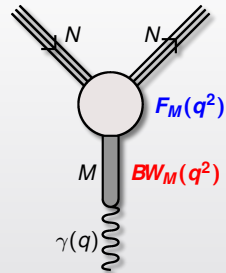


# **The Lomon-Gari-Krünpelmann Model**



The model is based on an idea of Iachello, Jackson and Landé, later developed by Gari and Krümpelmann, and Lomon. Nucleon EMFFs are parametrized using a **mixture** of Vector Meson Dominance (VMD) and pQCD.

- At low energy the coupling to the photons is described through vector meson exchange [VMD in terms of **propagators**  $BW_M(q^2)$ ,  $M = \rho, \omega, \phi, \rho', \omega'$ ].
- Hadron/quark form factors**  $F_M(q^2)$  at vector meson-nucleon (quark) vertices control transition to perturbative QCD at high momentum transfer.



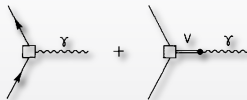
## Analytic extension: space-like $\longrightarrow$ time-like

- $BW_M(q^2)$  for broad mesons:**  
simple poles  $\longrightarrow$  poles with finite energy-dependent widths
- Dispersion relations:**  
rigorous analytic continuation of  $BW_M(q^2)$  from time-like to space-like region

# Parametrization for the isospin components

The **VMD** parametrization is for isospin components to single out different species of vector meson contributions

Isoscalar:  $\omega, \omega', \phi$     Isovector:  $\rho, \rho'$



## Isospin components of Dirac and Pauli EMFFs

$$F_1^{iv}(q^2) = [BW_\rho(q^2) + BW_{\rho'}(q^2)] F_1^\rho(q^2) + [1 - BW_\rho(0) - BW_{\rho'}(0)] F_1^D(q^2)$$

$$F_2^{iv}(q^2) = [\kappa_\rho BW_\rho(q^2) + \kappa_{\rho'} BW_{\rho'}(q^2)] F_2^\rho(q^2) + [\kappa_{iv} - \kappa_\rho BW_\rho(0) - \kappa_{\rho'} BW_{\rho'}(0)] F_2^D(q^2)$$

$$F_1^{is}(q^2) = [BW_\omega(q^2) + BW_{\omega'}(q^2)] F_1^\omega(q^2) + BW_\phi(q^2) F_1^\phi(q^2) + [1 - BW_\omega(0) - BW_{\omega'}(0)] F_1^D(q^2)$$

$$F_2^{is}(q^2) = [\kappa_\omega BW_\omega(q^2) + \kappa_{\omega'} BW_{\omega'}(q^2)] F_2^\omega(q^2) + \kappa_\phi BW_\phi(q^2) F_2^\phi(q^2) + [\kappa_{is} - \kappa_\omega BW_\omega(0) - \kappa_{\omega'} BW_{\omega'}(0)] F_2^D(q^2)$$

# The ingredients...

$$F_i^I(q^2) = \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(q^2) F_i^{V_1}(q^2) + \left[ \kappa_i^{i-1} - \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(0) F_i^{V_1}(0) \right] F_i^D(q^2)$$

$$BW_{V_1}(q^2) = \frac{g_{V_1} M_{V_1}^2}{f_{V_1}} \times [V_1\text{-propagator}] \begin{cases} (M_{V_1}^2 - q^2)^{-1} & V_1 = \omega, \phi \\ [\text{analytic}] & V_1 = \rho, \rho', \omega' \end{cases}$$

## Meson nucleon FFs

$$V_1 = \rho, \rho', \omega, \omega'$$

$$F_i^{V_1}(q^2) = f_i(q^2) = \left( \frac{\Lambda_1^2}{\Lambda_1^2 - q^2} \right) \left( \frac{\Lambda_2^2}{\Lambda_2^2 - q^2} \right)^i$$

$$F_1^\phi(q^2) = f_1(q^2) \left( \frac{\tilde{q}^2}{q^2 - \Lambda_1^2} \right)^{3/2}$$

$$F_2^\phi(q^2) = f_2(q^2) \left( \frac{\Lambda_1^2}{\mu_\phi^2} \frac{\tilde{q}^2 - \mu_\phi^2}{q^2 - \Lambda_1^2} \right)^{3/2}$$

## Photon-valence quark coupling

$$F_I^D(q^2) = \frac{\Lambda_D^2}{\Lambda_D^2 - q^2} \left( \frac{\Lambda_2^2}{\Lambda_2^2 - q^2} \right)^I$$

**QCD  
correction**

$$\tilde{q}^2 = q^2 \frac{\ln[(\Lambda_D^2 - q^2)/\Lambda_{\text{QCD}}]}{\ln(\Lambda_D^2/\Lambda_{\text{QCD}}^2)}$$

$$\kappa_{V_1} = \kappa_I$$

●  $\kappa_{V_1}$  is the ratio of tensor to vector  $NNV_1$ -coupling at  $q^2 = 0$

● Isospin anomalous magnetic moments:  $\begin{cases} \kappa_{\text{is}} = \kappa_p + \kappa_n \\ \kappa_{\text{iv}} = \kappa_p - \kappa_n \end{cases}$



# The ingredients...

$$F_i^I(q^2) = \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(q^2) F_i^{V_1}(q^2) + \left[ \kappa_i^{i-1} - \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(0) F_i^{V_1}(0) \right] F_i^D(q^2)$$

$$BW_{V_1}(q^2) = \frac{g_{V_1} M_{V_1}^2}{f_{V_1}} \times [V_1\text{-propagator}] \begin{cases} (M_{V_1}^2 - q^2)^{-1} & V_1 = \omega, \phi \\ [\text{analytic}] & V_1 = \rho, \rho', \omega' \end{cases}$$

## Meson nucleon FFs

$$V_1 = \rho, \rho', \omega, \omega'$$

$$F_i^{V_1}(q^2) = f_i(q^2) = \left( \frac{\Lambda_1^2}{\Lambda_1^2 - q^2} \right) \left( \frac{\Lambda_2^2}{\Lambda_2^2 - q^2} \right)^i$$

$$F_1^\phi(q^2) = f_1(q^2) \left( \frac{\tilde{q}^2}{q^2 - \Lambda_1^2} \right)^{3/2}$$

$$F_2^\phi(q^2) = f_2(q^2) \left( \frac{\Lambda_1^2}{\mu_\phi^2} \frac{\tilde{q}^2 - \mu_\phi^2}{q^2 - \Lambda_1^2} \right)^{3/2}$$

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QCD  
correction

$$\tilde{q}^2 = q^2 \frac{\ln[(\Lambda_D^2 - q^2)/\Lambda_{\text{QCD}}]}{\ln(\Lambda_D^2/\Lambda_{\text{QCD}}^2)}$$

$$\kappa_{V_1} = \kappa_1$$

●  $\kappa_{V_1}$  is the ratio of tensor to vector  $NNV_1$ -coupling at  $q^2 = 0$

● Isospin anomalous magnetic moments:  $\begin{cases} \kappa_{1s} = \kappa_p + \kappa_n \\ \kappa_{1v} = \kappa_p - \kappa_n \end{cases}$



# The ingredients...

$$F_i^I(q^2) = \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(q^2) F_i^{V_1}(q^2) + \left[ \kappa_1^{i-1} - \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(0) F_i^{V_1}(0) \right] F_i^D(q^2)$$

$$BW_{V_1}(q^2) = \frac{g_{V_1} M_{V_1}^2}{f_{V_1}} \times [V_1\text{-propagator}] \begin{cases} (M_{V_1}^2 - q^2)^{-1} & V_1 = \omega, \phi \\ [\text{analytic}] & V_1 = \rho, \rho', \omega' \end{cases}$$

## Meson nucleon FFs

$$V_1 = \rho, \rho', \omega, \omega'$$

$$F_i^{V_1}(q^2) = f_i(q^2) = \left( \frac{\Lambda_1^2}{\Lambda_1^2 - \tilde{q}^2} \right) \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

$$F_1^\phi(q^2) = f_1(q^2) \left( \frac{\tilde{q}^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

$$F_2^\phi(q^2) = f_2(q^2) \left( \frac{\Lambda_1^2}{\mu_\phi^2} \frac{\tilde{q}^2 - \mu_\phi^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

## Photon-valence quark coupling

$$F_1^D(q^2) = \frac{\Lambda_D^2}{\Lambda_D^2 - \tilde{q}^2} \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

**QCD  
correction**

$$\tilde{q}^2 = q^2 \frac{\ln[(\Lambda_D^2 - q^2)/\Lambda_{\text{QCD}}]}{\ln(\Lambda_D^2/\Lambda_{\text{QCD}}^2)}$$

$$\kappa_{V_1} \quad \kappa_1$$

- $\kappa_{V_1}$  is the ratio of tensor to vector  $NNV_1$ -coupling at  $q^2 = 0$
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# The ingredients...

$$F_i^I(q^2) = \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(q^2) F_i^{V_1}(q^2) + \left[ \kappa_1^{i-1} - \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(0) F_i^{V_1}(0) \right] F_i^D(q^2)$$

$$BW_{V_1}(q^2) = \frac{g_{V_1} M_{V_1}^2}{f_{V_1}} \times [V_1\text{-propagator}] \begin{cases} (M_{V_1}^2 - q^2)^{-1} & V_1 = \omega, \phi \\ [\text{analytic}] & V_1 = \rho, \rho', \omega' \end{cases}$$

Meson nucleon FFs

$$V_1 = \rho, \rho', \omega, \omega'$$

$$F_i^{V_1}(q^2) = f_i(q^2) = \left( \frac{\Lambda_1^2}{\Lambda_1^2 - \tilde{q}^2} \right) \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

$$F_1^\phi(q^2) = f_1(q^2) \left( \frac{\tilde{q}^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

$$F_2^\phi(q^2) = f_2(q^2) \left( \frac{\Lambda_1^2}{\mu_\phi^2} \frac{\tilde{q}^2 - \mu_\phi^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

Photon-valence quark coupling

$$F_i^D(q^2) = \frac{\Lambda_D^2}{\Lambda_D^2 - \tilde{q}^2} \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

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# The ingredients...

$$F_i^I(q^2) = \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(q^2) F_i^{V_1}(q^2) + \left[ \kappa_1^{i-1} - \sum_{V_1} \kappa_{V_1}^{i-1} BW_{V_1}(0) F_i^{V_1}(0) \right] F_i^D(q^2)$$

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Meson nucleon FFs

$$V_1 = \rho, \rho', \omega, \omega'$$

$$F_i^{V_1}(q^2) = f_i(q^2) = \left( \frac{\Lambda_1^2}{\Lambda_1^2 - \tilde{q}^2} \right) \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

$$F_1^\phi(q^2) = f_1(q^2) \left( \frac{\tilde{q}^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

$$F_2^\phi(q^2) = f_2(q^2) \left( \frac{\Lambda_1^2}{\mu_\phi^2} \frac{\tilde{q}^2 - \mu_\phi^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

Photon-valence quark coupling

$$F_i^D(q^2) = \frac{\Lambda_D^2}{\Lambda_D^2 - \tilde{q}^2} \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

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# The ingredients...

$$F_i^l(q^2) = \sum_{V_l} \kappa_{V_l}^{i-1} BW_{V_l}(q^2) F_i^{V_l}(q^2) + \left[ \kappa_l^{i-1} - \sum_{V_l} \kappa_{V_l}^{i-1} BW_{V_l}(0) F_i^{V_l}(0) \right] F_i^D(q^2)$$

$$BW_{V_l}(q^2) = \frac{g_{V_l} M_{V_l}^2}{f_{V_l}} \times [V_l\text{-propagator}] \begin{cases} (M_{V_l}^2 - q^2)^{-1} & V_l = \omega, \phi \\ [\text{analytic}] & V_l = \rho, \rho', \omega' \end{cases}$$

Meson nucleon FFs

$$V_l = \rho, \rho', \omega, \omega'$$

$$F_i^{V_l}(q^2) = f_i(q^2) = \left( \frac{\Lambda_1^2}{\Lambda_1^2 - \tilde{q}^2} \right) \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

$$F_1^\phi(q^2) = f_1(q^2) \left( \frac{\tilde{q}^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

$$F_2^\phi(q^2) = f_2(q^2) \left( \frac{\Lambda_1^2}{\mu_\phi^2} \frac{\tilde{q}^2 - \mu_\phi^2}{\tilde{q}^2 - \Lambda_1^2} \right)^{3/2}$$

Photon-valence quark coupling

$$F_i^D(q^2) = \frac{\Lambda_D^2}{\Lambda_D^2 - \tilde{q}^2} \left( \frac{\Lambda_2^2}{\Lambda_2^2 - \tilde{q}^2} \right)^i$$

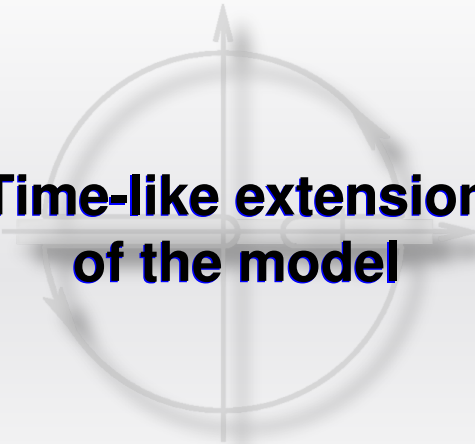
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$$\kappa_{V_l} \quad \kappa_l$$

- $\kappa_{V_l}$  is the ratio of tensor to vector  $NNV_l$ -coupling at  $q^2 = 0$
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# **Time-like extension of the model**

# Analytic Breit-Wigner formulas

Relativistic Breit-Wigner formula for an unstable particle with mass  $M$  and constant width  $\Gamma$

$$BW(s) = \frac{1}{M^2 - s - i\Gamma M}$$

In the  $s$ -complex plane

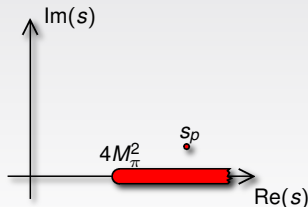
- Single pole
- No discontinuity cut

Energy-dependent widths are introduced considering decay rates extended to off shell particle masses. In the  $\rho$  meson case with  $\Gamma(\rho \rightarrow \pi^+\pi^-) = \Gamma_\rho$ .

$$\Gamma(s) = \frac{\Gamma_\rho^0 (s - 4M_\pi^2)^{3/2}}{M_\rho s}$$

$$BW(s) = \frac{s}{s(M_\rho^2 - s) - i\Gamma_\rho^0 (s - 4M_\pi^2)^{3/2}}$$

- Has the "required" discontinuity cut  $(4M_\pi^2, \infty)$
- Maintains a complex pole  $s_\rho \simeq M_\rho^2 + i\Gamma_\rho M_\rho$ , slightly shifted w.r.t. the original position
- The power "3/2" in the denominator and the factor  $1/s$  generate additional "physical" poles



# Regularization of Breit-Wigner formulas

The BW formula with energy-dependent width has a set of  $N$  poles  $\{z_j\}_{j=1}^N$  in the  $s$ -complex plane

- $P_N(s)$  is a suitable  $N$  degree polynomial
- $\beta$  is a noninteger real number which defines the discontinuity cut

$$BW(s) = \frac{P_N(s) \prod_{j=1}^N (s - z_j)^{-1}}{M_\rho^2 - s - i\gamma(s - 4M_\pi^2)^\beta}$$

To avoid unphysical divergences poles must be subtracted. BW formulas are **regularized** by adding counterparts that behave as the opposite of each pole.

## Method #1

The subtraction can be done by hand. . .

$$\widetilde{BW}(s) = BW(s) - \sum_{k=1}^N \frac{P_N(z_k) \prod_{j=1, j \neq k}^N (z_k - z_j)^{-1}}{M_\rho^2 - z_k - i\gamma(z_k - 4M_\pi^2)^\beta} \times \frac{1}{s - z_k}$$

- **Advantage:** easy to handle and to implement in codes
- **Drawback:** we need to know the pole positions

# Regularization using dispersion relations

If  $f(z)$  is an analytic function in the whole  $z$  complex plane with a real positive cut  $(s_0, \infty)$  and  $f(z) = o(1/\ln|z|)$  as  $z \rightarrow \infty$  then ....

$$f(z) = \frac{1}{\pi} \int_{s_0}^{\infty} \frac{\text{Im}[f(x)] dx}{x - z}$$

If the function  $f(z)$  has also a finite number of isolated poles  $\{z_j\}_{j=1}^N$ , **residues** have to be considered.....

$$f(z) + 2\pi i \sum_{j=1}^N \text{Res} \left[ \frac{f(z')}{z' - z}, z_j \right] = \frac{1}{\pi} \int_{s_0}^{\infty} \frac{\text{Im}[f(x)] dx}{x - z}$$

Writing

$$f(z) = \phi(z) \prod_{j=1}^N \frac{1}{z - z_j}$$

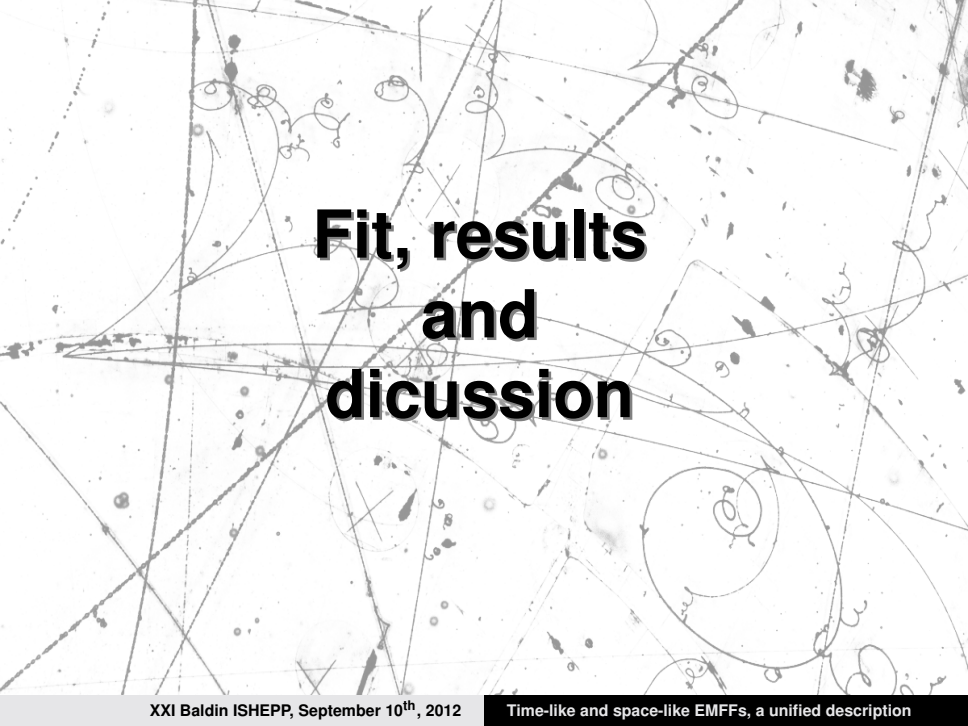
$\phi(z)$  is the pole-free part of  $f(z)$

$$f(z) + \sum_{k=1}^N \frac{\phi(z_k)}{z_k - z} \prod_{j \neq k}^N \frac{1}{z_k - z_j} = \frac{1}{\pi} \int_{s_0}^{\infty} \frac{\text{Im}[f(x)] dx}{x - z}$$

## Method #2

$$\widetilde{BW}(s) = \frac{1}{\pi} \int_{4M^2}^{\infty} \frac{\text{Im}[BW(s')]}{s' - s} ds'$$

- **Advantage:** we do not need to know the coordinates of the poles
- **Drawback:** we have to compute integrals over infinite intervals

The background of the slide is a complex, abstract pattern of thin black lines and small dots on a white background. The lines are mostly straight and intersect at various angles, creating a web-like structure. There are also several curved lines, some forming loops or spirals. The dots are scattered throughout, often appearing along the lines or in clusters. The overall effect is that of a technical drawing or a scientific sketch.

# **Fit, results and dicussion**

# The fitting procedure

## Space-like region

| Quantity ( $Q_i$ )     | $G_M^p$ | $G_E^p$ | $G_M^n$ | $G_E^n$ | $\frac{\mu_p G_E^p}{G_M^p}$ | $\frac{\mu_n G_E^n}{G_M^n}$ |
|------------------------|---------|---------|---------|---------|-----------------------------|-----------------------------|
| n. of points ( $N_i$ ) | 68      | 36      | 65      | 14      | 25                          | 13                          |

## Time-like

| $ G_{\text{eff}}^p $ | $ G_{\text{eff}}^n $ |
|----------------------|----------------------|
| 81(43)               | 5                    |

Global  $\chi^2$

$$\chi^2 = \sum_{i=1}^9 \tau_i \cdot \chi_i^2$$

Contribution of the data set:  $\{q_k^2, v_k^i, \delta v_k^i; N_i\}$

$$\chi_i^2 = \sum_{k=1}^{N_i} \left( \frac{Q_i(q_k^2) - v_k^i}{\delta v_k^i} \right)^2$$

## 13 Free parameters

$\Lambda_1, \Lambda_2, \Lambda_D$

Parametrize hadronic FFs and control the transition from non perturbative to perturbative QCD

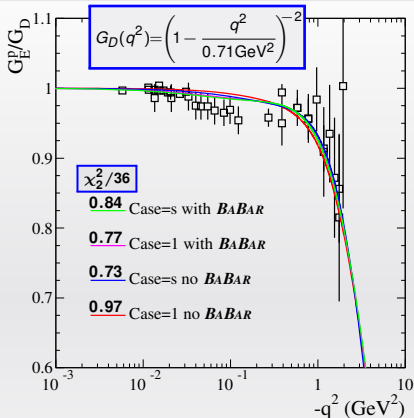
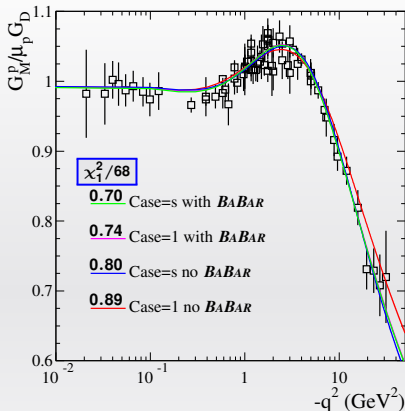
Five pairs ( $\kappa_{V_1}, g_{V_1}/f_{V_1}$ ) of vector meson anomalous magnetic momenta and couplings with:

$$V_1 = \rho, \rho', \omega, \omega', \phi$$

- Masses and widths of all vector mesons are fixed to the PDG values
- For the QCD scale two values have been considered:  $\Lambda_{\text{QCD}} = 0.15, 0.10 \text{ GeV}$

# Proton space-like $G_E^p$ and $G_M^p$

Data collection:  
PRC82,045211



## Two BW analytic formulas

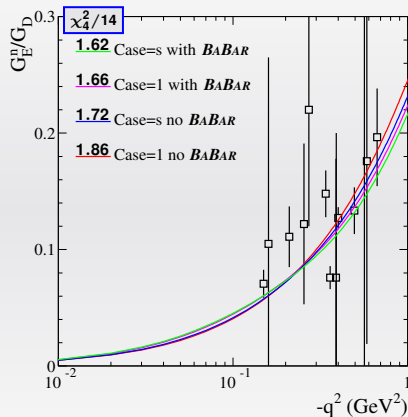
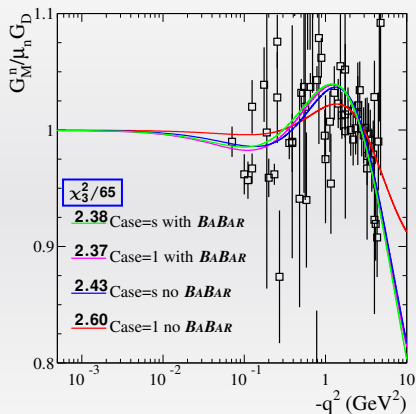
- **Case=s:**  $\Gamma_s(s) = \frac{\Gamma_s^0}{M} \frac{(s - s_{\text{th}})^{3/2}}{s}$
- **Case=1:**  $\Gamma_1(s) = \frac{\Gamma_1^0}{M^3} (s - s_{\text{th}})^{3/2}$

**Two data sets:** including and not *BABAR* time-like data on  $|G_{\text{eff}}^p(q^2)|$  extracted from the radiative process



- with *BABAR*
- no *BABAR*

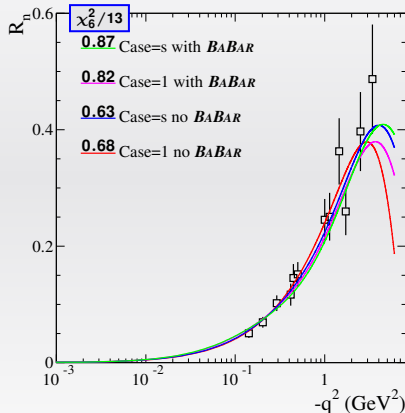
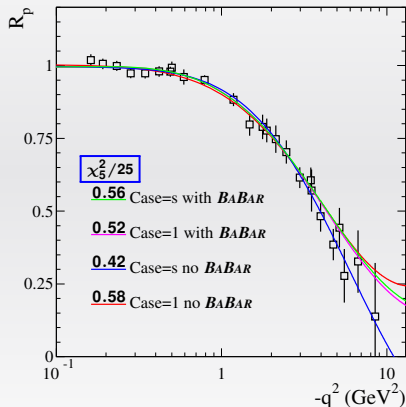




# Space-like ratios

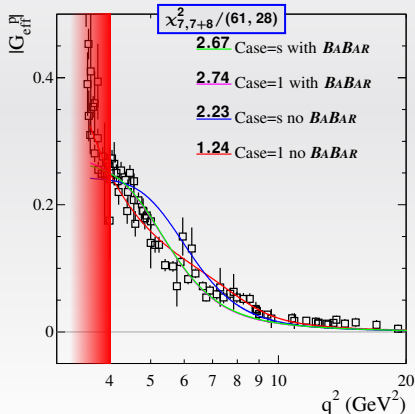
$$R_p = \mu_p G_E^p / G_M^p \text{ and } R_n = \mu_n G_M^n / G_M^n$$

Data collection:  
PRC82,045211



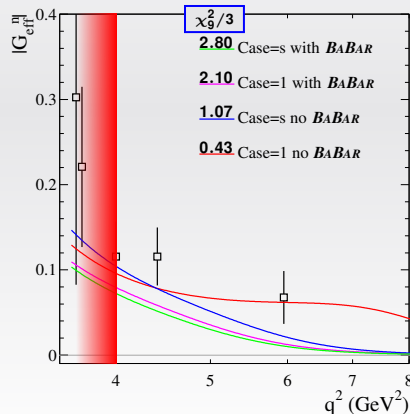
## From experiments

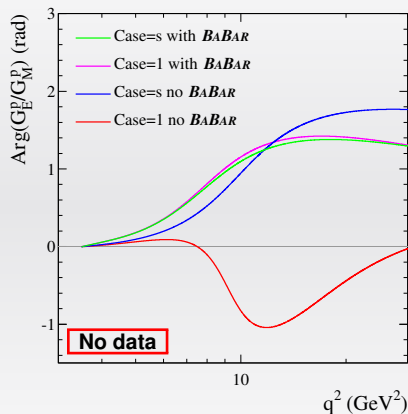
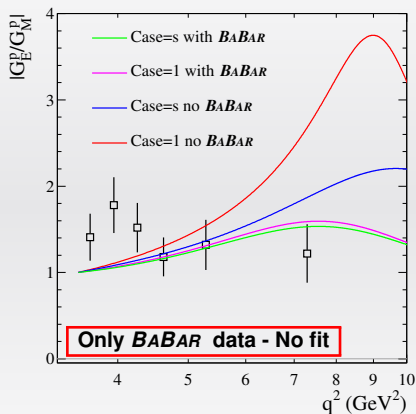
$$|G_{\text{eff}}^N| = \sqrt{\frac{\sigma(e^+e^- \rightarrow N\bar{N})}{\frac{4\pi\alpha^2}{3q^2} \sqrt{1 - \frac{4M_N^2}{q^2}} \left(1 + \frac{2M_N^2}{q^2}\right)}}$$



## Fitting function

$$|G_{\text{eff}}^N| = \sqrt{\frac{|G_M^N(q^2)|^2 + \frac{2M_N^2}{q^2} |G_E^N(q^2)|^2}{1 + \frac{2M_N^2}{q^2}}}$$





# Summary of $\chi^2$ contributions

|                     | $Q_i$                 | $N_i$    | minimum $\chi_i^2$             |                             |                           |                           |
|---------------------|-----------------------|----------|--------------------------------|-----------------------------|---------------------------|---------------------------|
|                     |                       |          | case=s<br>With<br><i>BABAR</i> | case=1<br>With <i>BABAR</i> | case=s<br>No <i>BABAR</i> | case=1<br>No <i>BABAR</i> |
| space-like          | $G_M^p$               | 68       | 48.7                           | 50.1                        | 54.6                      | 60.8                      |
|                     | $G_E^p$               | 36       | 30.4                           | 27.6                        | 26.2                      | 35.0                      |
|                     | $G_M^n$               | 65       | 154.6                          | 154.2                       | 158.2                     | 167.0                     |
|                     | $G_E^n$               | 14       | 22.7                           | 23.2                        | 24.1                      | 26.0                      |
|                     | $\mu_p G_E^p / G_M^p$ | 25       | 13.9                           | 12.9                        | 10.6                      | 14.4                      |
|                     | $\mu_n G_E^n / G_M^n$ | 13       | 11.3                           | 10.7                        | 8.2                       | 8.9                       |
| time-like           | $ G_{\text{eff}}^p $  | 61 (28)  | 162.5                          | 166.7                       | 62.2                      | 35.0                      |
|                     | $ G_{\text{eff}}^n $  | 3        | 8.4                            | 6.3                         | 3.2                       | 0.3                       |
| Total               |                       | 285(252) | 452.5                          | 451.7                       | 347.3                     | 347.4                     |
| Normalized $\chi^2$ |                       |          | 1.663                          | 1.661                       | 1.453                     | 1.454                     |

# Best values of fit parameters and constants

| Parameter                    | case = s<br>With <i>BABAR</i> | case = 1<br>With <i>BABAR</i> | case = s<br>No <i>BABAR</i> | case = 1<br>No <i>BABAR</i> |
|------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|
| $g_\rho / f_\rho$            | 2.766                         | 2.410                         | 0.9029                      | 0.4181                      |
| $\kappa_\rho$                | -1.194                        | -1.084                        | 0.8267                      | 0.6885                      |
| $M_\rho$ (GeV)               | 0.7755 (fixed)                |                               |                             |                             |
| $\Gamma_\rho$ (GeV)          | 0.1491 (fixed)                |                               |                             |                             |
| $g_\omega / f_\omega$        | -1.057                        | -1.043                        | -0.2308                     | -0.4894                     |
| $\kappa_\omega$              | -3.240                        | -3.317                        | -9.859                      | -1.398                      |
| $M_\omega$ (GeV)             | 0.78263 (fixed)               |                               |                             |                             |
| $g_\phi / f_\phi$            | 0.1871                        | 0.1445                        | -0.0131                     | -0.1156                     |
| $\kappa_\phi$                | -2.004                        | -3.045                        | 37.218                      | -0.2613                     |
| $M_\phi$ (GeV)               | 1.019 (fixed)                 |                               |                             |                             |
| $\mu_\phi$ (GeV)             | 20.0 (fixed)                  |                               |                             |                             |
| $g_{\omega'} / f_{\omega'}$  | 2.015                         | 1.974                         | 1.265                       | 1.649                       |
| $\kappa_{\omega'}$           | -2.053                        | -2.010                        | -2.044                      | -0.6712                     |
| $M_{\omega'}$ (GeV)          | 1.425 (fixed)                 |                               |                             |                             |
| $\Gamma_{\omega'}$ (GeV)     | 0.215 (fixed)                 |                               |                             |                             |
| $g_{\rho'} / f_{\rho'}$      | -3.475                        | -3.274                        | -0.8730                     | -0.0369                     |
| $\kappa_{\rho'}$             | -1.657                        | -1.724                        | -2.832                      | -104.35                     |
| $M_{\rho'}$ (GeV)            | 1.465 (fixed)                 |                               |                             |                             |
| $\Gamma_{\rho'}$ (GeV)       | 0.400 (fixed)                 |                               |                             |                             |
| $\Lambda_1$ (GeV)            | 0.4801                        | 0.5000                        | 0.6474                      | 0.6446                      |
| $\Lambda_2$ (GeV)            | 3.0536                        | 3.0562                        | 3.0872                      | 3.6719                      |
| $\Lambda_D$ (GeV)            | 0.7263                        | 0.7416                        | 0.8573                      | 0.8967                      |
| $\Lambda_{\text{QCD}}$ (GeV) | 0.150                         |                               |                             |                             |

## Time-like extension of the Lomon-Gari-Krümpelman model

- Breit-Wigner formulas describing broad intermediate vector mesons have been modified including energy-dependent widths in two scenarios:  $\Gamma_1(q^2)$ , minimal alteration, and  $\Gamma_s(q^2)$  derived from relativistic perturbation theory
- A **regularization procedure** has been defined to remove unwanted poles and so to fulfill the analyticity requirements in the whole  $q^2$  complex plane

## Fit results

- The improvements in the BW formulas do not affect the space-like fit quality
- The **simultaneous space-like and time-like** fit is satisfactory
- The  $\chi^2$  contributions from each space-like data set are almost **unchanged** between **case=1** and **case=s**
- The quality of the fit is **poorer when BABAR data are included**

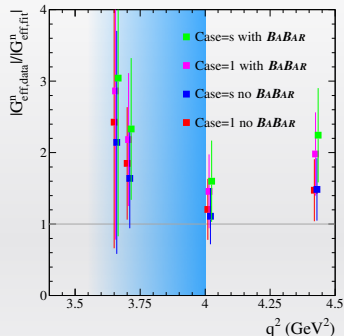
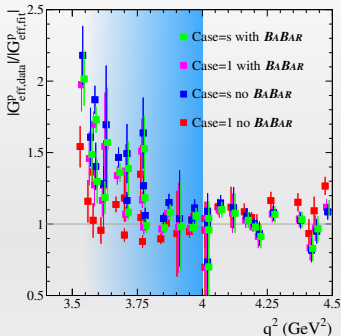
## Predictions

- The knowledge of the analytic structure of EMFFs allows us to make predictions on **polarization observables** like EMFF **phases** and “**pure**” **moduli**
- Measurements of such observables would be effective in **discriminating among the different models and parametrizations**

# BACK-UP SLIDES

# Time-like threshold behavior

$$|G_{\text{eff}}^N| = \sqrt{\sigma(e^+e^- \rightarrow N\bar{N})} \left[ \frac{4\pi\alpha^2}{3q^2} \sqrt{1-4M_N^2/q^2} (1+2M_N^2/q^2) \right]^{-1/2}$$



- As a consequence of near-threshold **flat cross sections**, effective nucleon EMFFs have a steep enhancement when  $q^2 \rightarrow (2M_N)^2$
- Such a flat cross section is in contrast with the expectation in case of smooth EMFFs ( $\sigma \propto \beta_N$ )
- To extract Born cross section and hence  $|G_{\text{eff}}^N|$  in the threshold region, data have to be corrected for **Coulomb** as well as **strong** effects

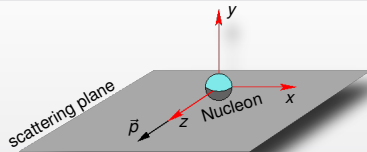
To avoid ambiguities due to the not well known form and interplay of these threshold corrections, **time-like data below  $q^2 = 4 \text{ GeV}^2$  have not been considered**

# Polarization formulae in the time-like region

The ratio  $R(q^2)$  is complex for  $q^2 \geq s_{th}$

$$R(q^2) = \mu_p \frac{G_E^p(q^2)}{G_M^p(q^2)} = |R(q^2)| e^{i\rho(q^2)}$$

The polarization depends on the phase  $\rho$



[A.Z. Dubnickova, S. Dubnicka, M.P. Rekalo, NCA109,241(96)]

$$\mathcal{P}_y = - \frac{\sin(2\theta)|R| \sin(\rho)}{D\sqrt{\tau}} = \frac{d\sigma^\uparrow - d\sigma^\downarrow}{d\sigma^\uparrow + d\sigma^\downarrow} \equiv \mathcal{A}_y \left\} \text{Does not depend on } P_e\right.$$

$$\mathcal{P}_x = - P_e \frac{2 \sin(2\theta)|R| \cos(\rho)}{D\sqrt{\tau}}$$

$$\mathcal{P}_z = P_e \frac{2 \cos(\theta)}{D} \left\} \text{Does not depend on } \rho\right.$$

$$D = \frac{1 + \cos^2 \theta + \frac{1}{\tau} |R|^2 \sin^2 \theta}{\mu_p}, \quad \tau = \frac{q^2}{4M_N^2}, \quad P_e = \text{electron polarization}$$